Naima Azim

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Software Engineer with 2 years of experience in Python, Odoo ERP, REST APIs, and database-driven applications. MSc graduate in Computer Science and Engineering (Data Science), with research experience in machine learning, computer vision, and digital twins.

Technical Skills

Programming: Python, SQL, JavaScript, Java, C, C++, HTML, CSS
Backend & ERP: Odoo, Odoo ORM, Custom Module Development, REST APIs, CRUD Operations, Authentication, Authorization
Databases & Reporting: PostgreSQL, MySQL, Complex SQL Queries, Database Models and Relationships, PDF/Excel Reports
AI/ML & Data Science: TensorFlow, YOLOv5, Physics-Informed Machine Learning, Feature Engineering, Model Evaluation, Data Analysis
Frontend & Visualization: XML Views, Odoo Widgets, HTML Templates, CSS, JavaScript, WebSockets, Basic React.js
Tools & Platforms: Git, PyCharm, PgAdmin, Google Colab, Android Studio, Unity, Visual Studio

Work Experience

Junior Software Engineer *Sep 2024 – Aug 2026*
Unisoft Systems Limited

- Designed and developed approximately six custom Odoo ERP modules using Python, PostgreSQL, SQL, XML, and the Odoo ORM, implementing business workflows, computed fields, server actions, custom views, and access controls.
- Developed REST API endpoints for web and mobile applications, implementing CRUD operations, token-based authentication, user authorization, input validation, and error handling.
- Built a customer-facing ticketing system with website-integrated APIs, template-based interfaces, JavaScript functionality, and role-based access controls for engineers, service managers, and help desk teams.
- Developed sales target distribution modules using accounting transaction data to track salesperson performance against daily and monthly targets. Created interactive dashboards and organizational hierarchy visualizations using JavaScript, Odoo widgets, HTML templates, and CSS.
- Implemented a technician commission management module with web and mobile API integrations, security groups, and customized reports.
- Wrote complex SQL queries for accounting and inventory reports, modified database models and relationships, and generated custom PDF and Excel reports.
- Contributed hands-on to ERP upgrades from Odoo 13 to 15 and Odoo 15 to 18 by modifying existing modules, resolving compatibility issues, and testing selected modules.
- Diagnosed production errors and reporting issues, resolving straightforward problems within hours and more complex issues within two working days. Implemented and tested enhancements to existing live systems within timelines of up to seven days.
- Managed selected projects independently while collaborating with developers and communicating with clients to clarify requirements, resolve issues, document technical changes, and deliver features.
- Delivered client training on system workflows and functionality to help users understand and effectively use their ERP solutions.

Research Projects

Physics-Informed Gearbox Fault Diagnosis and Digital Twin *MSc Thesis Research*
Python, TensorFlow, Machine Learning, Flask, JavaScript, WebSockets, Unity 3D

- Developed a machine-learning-based gearbox fault diagnosis approach using simulated gearbox vibration data and an 18-dimensional physics-informed feature set.
- Incorporated physics-based equations and baseline values into feature engineering to represent relevant gearbox characteristics for fault diagnosis.
- Evaluated more than 15 machine-learning models using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices, comparing results through performance visualizations.

- Achieved 99.26% accuracy and 0.99 ROC-AUC with the selected model on the simulated dataset.
- Developed an interactive digital twin dashboard using Flask, JavaScript, and WebSockets, integrated with a Unity 3D gearbox simulation for live machine-condition and fault visualization.

Sign Language Recognition Using Deep Learning *BSc Research Project*
YOLOv5, Computer Vision, Android Studio

- Developed an Android application for real-time sign language recognition using a custom-trained YOLOv5 object detection model.
- Worked with image data and CSV-based data resources for the sign language recognition project.
- Integrated the recognition model into an Android application and developed a multilingual sign language dictionary as an in-app reference feature.

Publication

Azim, N., AfrinJamema, S., and Sultana, N. (2024). "Study and Analysis of Deep Learning Models for the Recognition of Sign Language." *Medicon Engineering Theme*, 6(1), 22–32.

Education

MSc in Computer Science & Engineering *2025 – 2026*
Daffodil International University, Bangladesh

Major: Data Science GPA: 3.93/4.00

BSc in Computer Science & Engineering *2018 – 2022*
Daffodil International University, Bangladesh

GPA: 3.90/4.00 Graduated in the top 2% of the class

Additional Experience

Teacher – Mathematics & Computer Science *Mar 2023 – Jun 2024*
Radius International School

Taught Mathematics and introductory Python to primary-level students, adapting lessons to different learning needs.

Additional Information

English Proficiency: IELTS Academic – Overall Band 7.5/9.0
Languages: Bengali (Native), English (Fluent), French (Basic; learning)
Interests: Applied AI/ML, Intelligent Systems, Digital Twins, Computer Vision, Software Development